

CSCE 222

Homework 5

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Contents

0.1	12	2
0.1.1	a	2
0.1.2	b	3
0.1.3	c	3
0.1.4	d	3
0.2	Question 14 - Find the recurrence relation satisfied by the sequence $\{a_n\}$	4
0.2.1	a	4
0.2.2	b	4
0.2.3	c	4
0.2.4	d	4
0.2.5	e	5
0.2.6	f	5
0.2.7	g	5
0.2.8	h	5
0.3	Question 22a - An employee joined a company in 2023 with a starting salary of \$50,000. Every year this employee receives a raise of \$1000 plus 5% of the salary of the previous year.	5
0.3.1	a	5

0.4	Question 22b - What will the salary of this employee be in 2031?	6
0.5	Question 32 - Find the value of each of these sums.	6
0.5.1	a	6
0.5.2	b	6
0.5.3	c	6
0.5.4	d	7
1	Section 5.1	7
1.1	Question 8 - Prove that $2 - 2 \cdot 7 + 2 \cdot 7^2 - \dots + 2(-7)^n = (1 - (-7)^{n+1})/4$ whenever n is a nonnegative integer.	7
1.2	Question 12 - Prove $\sum_{j=0}^n (-\frac{1}{2})^j = \frac{2^{n+1} + (-1)^n}{3 \cdot 2^n}$ for $n \geq 0$	8
1.3	Question 14 - Prove that for every positive integer n , $\sum_{k=1}^n (k2^k) = (n-1)2^{n+1} + 2$	8
1.4	Question 18 - Prove by mathematical induction: Let $P(n)$ be the statement that $n! < n^n$, where n is an integer greater than 1 . . .	9

0.1 12 - Show that the sequence $\{a_n\}$ is a solution of the recurrence relation $a_n = -3a_{n-1} + 4a_{n-2}$ if

$$P(n) = a_n = -3a_n - 1 + 4a_{n-2} \text{ for all } n \geq 2$$

0.1.1 a - $a_n = 0$

Basis step: $P(2) = a_2 = 0 = -3(0) + 4(0) = 0$ which holds

Inductive step: $p(k) \rightarrow p(k+1)$

Assume $p(k)$ holds for $k \geq 2$, $a_k = 0, a_{k-1} = 0$

$$\text{LHS: } a_{k+1} = 0$$

$$\text{RHS: } -3a_k + 4a_{k-1}$$

$$-3(0) + 4(0) = 0$$

LHS = RHS therefore $p(k) \rightarrow p(k+1)$ holds for all $n \geq 2$ by induction.

0.1.2 b - $a_n = 1$

Basis step: $P(2) = a_2 = 1 = -3(1) + 4(1) = 1$ which holds

Inductive step: $p(k) \rightarrow p(k+1)$

Assume $p(k)$ holds for $k \geq 2$, $a_k = 1, a_{k-1} = 1$

LHS: $a_{k+1} = 1$

RHS: $-3a_k + 4a_{k-1}$

$-3(1) + 4(1) = 1$

LHS = RHS therefore $p(k) \rightarrow p(k+1)$ holds for all $n \geq 2$ by induction.

0.1.3 c - $a_n = (-4)^n$

Basis step: $P(2) = a_2 = (-4)^2 = 16 = -3((-4)^1) + 4((-4)^0) = 12 + 4 = 16$
which holds

Inductive step: $p(k) \rightarrow p(k+1)$

Assume $p(k)$ holds for $k \geq 2$, $a_k = -4^k, a_{k-1} = -4^{k-1}$

LHS: $a_{k+1} = (-4)^{k+1}$

RHS: $-3a_k + 4a_{k-1}$

$-3((-4)^k) + 4((-4)^{k-1})$

$-3((-4)^k) + 4((-4)^k(-4)^{-1})$

$-3((-4)^k) + ((-4)^k(-1))$

$-4(-4)^k$

$(-4)^1((-4)^k)$

$((-4)^{k+1})$

LHS = RHS therefore $p(k) \rightarrow p(k+1)$ holds for all $n \geq 2$ by induction.

0.1.4 d - $a_n = 2(-4)^n + 3$

Basis step: $P(2) = a_2 = 2(-4)^2 + 3 = 35 = -3(2(-4)^1 + 3) + 4(2(-4)^0 + 3) =$
 $15 + 20 = 35$ which holds

Inductive step: $p(k) \rightarrow p(k+1)$

Assume $p(k)$ holds for $k \geq 2$, $a_k = 2(-4)^k + 3, a_{k-1} = 2(-4)^{k-1} + 3$

LHS: $2(-4)^{k+1} + 3$

$$\text{RHS: } -3a_k + 4a_{k-1}$$

$$-3(2(-4)^k + 3) + 4(2(-4)^{k-1} + 3)$$

$$(-6(-4)^k + -9) + (8(-4)^k(-4)^{-1} + 12)$$

$$(-6(-4)^k + -9) + ((-4)^k(-2) + 12)$$

$$(-6(-4)^k) + (-2(-4)^k) + 3$$

$$(-8(-4)^k) + 3$$

$$((2(-4)^1)(-4)^k) + 3$$

$$2(-4)^{k+1} + 3$$

LHS = RHS therefore $p(k) \rightarrow p(k+1)$ holds for all $n \geq 2$ by induction.

0.2 Question 14 - Find the recurrence relation satisfied by the sequence $\{a_n\}$

0.2.1 a - $a_n = 0$

$a_n = 0(a_{n-1})$ for all $n \geq 1$, where $a_0 = 0$

0.2.2 b - $a_n = 1$

$a_n = (a_{n-1})$ for all $n \geq 1$, where $a_0 = 1$

0.2.3 c - $a_n = 2^n$

$a_n = 2(a_{n-1})$ for all $n \geq 1$, where $a_0 = 1$

0.2.4 d - $a_n = 4^n$

$a_n = 4(a_{n-1})$ for all $n \geq 1$, where $a_0 = 1$

0.2.5 e - $a_n = n4^n$

$a_n = 8(a_{n-1}) - 16(a_{n-2})$ for all $n \geq 2$, where $a_0 = 0, a_1 = 4$

0.2.6 f - $a_n = 2(4^n) + 3n4^n$

$a_n = 8(a_{n-1}) - 16(a_{n-2})$ for all $n \geq 2$, where $a_0 = 2, a_1 = 20$

0.2.7 g - $a_n = (-4)^n$

$a_n = -4(a_{n-1})$ for all $n \geq 1$, where $a_0 = 1$

0.2.8 h - $a_n = n^2 4^n$

$a_n = 12(a_{n-1}) - 48(a_{n-2}) + 64(a_{n-3})$ for all $n \geq 3$, where $a_0 = 0, a_1 = 4, a_2 = 64$

0.3 Question 22a - An employee joined a company in 2023 with a starting salary of \$50,000. Every year this employee receives a raise of \$1000 plus 5% of the salary of the previous year.

0.3.1 a - Set up a recurrence relation for the salary of this employee n years after 2023.

$$a_0 = 50,000$$

$$a_n = a_{n-1} + 1,000 + 0.05a_{n-1}$$

$$a_n = 1.05a_{n-1} + 1,000 \text{ for all } n \geq 1, \text{ where } a_0 = 50,000$$

0.4 Question 22b - What will the salary of this employee be in 2031?

$$n = 2031 - 2023 = 8$$

$$a_n = 1.05a_{n-1} + 1,000 \text{ for all } n \geq 1, \text{ where } a_0 = 50,000$$

$$a_1 = 1.05a_0 + 1,000 = 53,500 \text{ in 2024}$$

$$a_2 = 1.05a_1 + 1,000 = 57,175 \text{ in 2025}$$

$$a_3 = 1.05a_2 + 1,000 = 61,033.75 \text{ in 2026}$$

$$a_4 = 1.05a_3 + 1,000 = 65,085.44 \text{ in 2027}$$

$$a_5 = 1.05a_4 + 1,000 = 69,339.71 \text{ in 2028}$$

$$a_6 = 1.05a_5 + 1,000 = 73,806.70 \text{ in 2029}$$

$$a_7 = 1.05a_6 + 1,000 = 78,497.04 \text{ in 2030}$$

$$a_8 = 1.05a_7 + 1,000 = 83,421.89 \text{ in 2031}$$

Thus, the salary in 2031 would be \$83,421.89

0.5 Question 32 - Find the value of each of these sums.

0.5.1 a- $\sum_{j=0}^8 1 + (-1)^j$

$$\sum_{j=0}^8 1 + (-1)^j = \sum_{j=0}^8 1 + \sum_{j=0}^8 (-1)^j$$

$$\sum_{j=0}^8 1 + (-1)^j = 9(1) + 1 + 4(-1 + 1) = 10$$

0.5.2 b- $\sum_{j=0}^8 (3^j - 2^j)$

$$\sum_{j=0}^8 (3^j - 2^j) = \sum_{j=0}^8 (3^j) - \sum_{j=0}^8 (2^j)$$

Using the geometric series formula:

$$\sum_{j=0}^8 (3^j - 2^j) = \left(\frac{3^0(1-3^9)}{1-3} \right) - \left(\frac{2^0(1-(2)^9)}{1-2} \right)$$

$$\sum_{j=0}^8 (3^j - 2^j) = (9841 - 511) = 9330$$

0.5.3 c- $\sum_{j=0}^8 (2 \cdot 3^j + 3 \cdot 2^j)$

Using the geometric series formula:

$$\sum_{j=0}^8 (2 \cdot 3^j + 3 \cdot 2^j) = 2 \left(\frac{3^0(1-3^9)}{1-3} \right) + 3 \left(\frac{2^0(1-(2)^9)}{1-2} \right)$$

using part b,

$$\sum_{j=0}^8 (2 \cdot 3^j + 3 \cdot 2^j) = 2(9841) + 3(511) = 21215$$

0.5.4 $\mathbf{d}\text{-}\sum_{j=0}^8 (2^{j+1} - 2^j)$

$$\sum_{j=0}^8 (2^{j+1} - 2^j) = \sum_{j=0}^8 (2^j 2 - 2^j)$$

$$\sum_{j=0}^8 (2^{j+1} - 2^j) = (2 \sum_{j=0}^8 2^j) - (\sum_{j=0}^8 (2^j)) = (\sum_{j=0}^8 (2^j))$$

Using the geometric series formula:

$$\sum_{j=0}^8 (2^{j+1} - 2^j) = \frac{1-2^9}{1-2} = 511$$

1 Section 5.1

1.1 Question 8 - Prove that $2 - 2 \cdot 7 + 2 \cdot 7^2 - \dots + 2(-7)^n = (1 - (-7)^{n+1})/4$ whenever n is a nonnegative integer.

$$P(n) = 2 - 2 \cdot 7 + 2 \cdot 7^2 - \dots + 2(-7)^n = (1 - (-7)^{n+1})/4 \text{ for all } n \geq 0$$

Basis step: $P(0) = a_0 = 2(-7)^0 = 2 = (1 - (-7)^{0+1})/4 = 2$ which holds

Inductive step: $p(k) \rightarrow p(k+1)$

Assume $p(k)$ holds for $k \geq 0$, $2 - 2 \cdot 7 + 2 \cdot 7^2 - \dots + 2(-7)^k = (1 - (-7)^{k+1})/4$

$$\text{RHS: } (1 - (-7)^{(k+1)+1})/4$$

$$\text{LHS: } (1 - (-7)^{k+1})/4 + a_{k+1}$$

$$(1 - (-7)^{k+1})/4 + 2(-7)^{k+1}$$

$$(1 - (-7)^{k+1})/4 + (8(-7)^{k+1})/4$$

$$(1 + (8 - 1)(-7)^{k+1})/4$$

$$(1 + 7(-7)^{k+1})/4$$

$$(1 - 7(-7)^{k+1})/4$$

$$(1 - (-7)^{(k+1)+1})/4$$

LHS = RHS therefore $p(k) \rightarrow p(k+1)$ holds for all $n \geq 0$ by induction.

1.2 Question 12 - Prove $\sum_{j=0}^n (-\frac{1}{2})^j = \frac{2^{n+1}+(-1)^n}{3 \cdot 2^n}$ for $n \geq 0$

$$P(n) = \sum_{j=0}^n (-\frac{1}{2})^j = \frac{2^{n+1}+(-1)^n}{3 \cdot 2^n} \text{ for all } n \geq 0$$

$$\text{Basis step: } P(0) = a_0 = \sum_{j=0}^0 (-\frac{1}{2})^0 = 1 = \frac{2^{0+1}+(-1)^0}{3 \cdot 2^0} = \frac{3}{3} = 1 \text{ which holds}$$

$$\text{Inductive step: } p(k) \rightarrow p(k+1)$$

$$\text{Assume } p(k) \text{ holds for } k \geq 0, \sum_{j=0}^k (-\frac{1}{2})^j = \frac{2^{k+1}+(-1)^k}{3 \cdot 2^k}$$

$$\text{RHS: } \frac{2^{(k+1)+1}+(-1)^{k+1}}{3 \cdot 2^{k+1}}$$

$$\text{LHS: } \sum_{j=0}^k (-\frac{1}{2})^j + (-\frac{1}{2})^{k+1}$$

$$\frac{2^{k+1}+(-1)^k}{3 \cdot 2^k} + (-\frac{1}{2})^{k+1}$$

$$\frac{2(2^{k+1}+(-1)^k)}{3 \cdot 2^{k+1}} + (\frac{3(-1)^{k+1}}{3 \cdot 2^{k+1}})$$

$$\frac{(2(2^{k+1})-(-1)^k)}{3 \cdot 2^{k+1}}$$

$$\frac{(2^{(k+1)+1}+(-1)^{k+1})}{3 \cdot 2^{k+1}}$$

$$\text{LHS} = \text{RHS} \text{ therefore } p(k) \rightarrow p(k+1) \text{ holds for all } n \geq 0 \text{ by induction.}$$

1.3 Question 14 - Prove that for every positive integer n ,

$$\sum_{k=1}^n (k2^k) = (n-1)2^{n+1} + 2$$

$$P(n) = \sum_{k=1}^n (k2^k) = (n-1)2^{n+1} + 2 \text{ for all } n \geq 1$$

$$\text{Basis step: } P(1) = a_1 = \sum_{k=1}^1 (k2^k) = 2 = (n-1)2^{n+1} + 2 = 0 + 2 \text{ which holds}$$

$$\text{Inductive step: } p(k) \rightarrow p(k+1)$$

$$\text{Assume } p(j) \text{ holds for } j \geq 1, \sum_{k=1}^j (k2^k) = (j-1)2^{j+1} + 2$$

$$\text{RHS: } ((j+1)-1)2^{(j+1)+1} + 2$$

$$(j)2^{j+2} + 2$$

$$\text{LHS: } \sum_{k=1}^j (k2^k) + (j+1)2^{j+1}$$

$$(j-1)2^{j+1} + 2 + (j+1)2^{j+1}$$

$$(2j)2^{j+1} + 2$$

$$(j)2^{j+2} + 2$$

$$\text{LHS} = \text{RHS} \text{ therefore } p(k) \rightarrow p(k+1) \text{ holds for all } n \geq 1 \text{ by induction.}$$

1.4 Question 18 - Prove by mathematical induction: Let $P(n)$ be the statement that $n! < n^n$, where n is an integer greater than 1

$P(n) = n! < n^n$ for all $n > 1$

a) Basis step: $P(2) = 2! < 2^2$

b) $P(2) = 2! = 2 < 2^2 = 4$ which holds

Inductive step: $p(k) \rightarrow p(k+1)$

c) Assume $p(k)$ holds for $k > 1$, $k! < k^k$

d) Need to prove that $p(k)$ implies $p(k+1)$ is true, meaning the LHS $\leq$ RHS for $p(k+1)$

e) RHS: $(k+1)^{k+1}$

$(k+1)(k+1)^k$

LHS: $(k+1)!$

$(k+1)(k)!$

$(k+1)(k)! < (k+1)(k^k)$ Application of the hypothesis

$(k+1)(k)! < (k+1)(k^k)$

Comparing LHS and RHS:

$(k+1)(k)! < (k+1)(k)^k < (k+1)(k+1)^k$ because $k < k+1$

f) These steps show under mathematical induction that starting with the smallest possible integer value for n greater than 1, (that is 2), the inequality holds. For any integer greater than or equal to 2 the inequality will continue to hold, as one integer implies the next holds the inequality.